until the 19th century, however, that the potential of such surveys was really understood.

The new study of ecology

As naturalists undertook longer field trips, the global distribution of species became more apparent, and the concept of ecology as a science gained traction.

One of the first scientists to employ ecological methods was American biologist Stephen A. Forbes. In the 1880s, while studying fish in a Wisconsin lake, he realized that survey data could be interpreted to give a picture of interactions between different species—not just their abundance. Forbes extended the scope of the conventional survey, combining practical fieldwork with theoretical analysis and experiments. These rounded ecological surveys created a picture of the natural order within an environment. By shedding light on the interrelated effects of its plant and animal life, they could also help explain the distribution of species and variations over time.